# Cantheria

Swarm Alignment Hackathon submission · Track 4: Multi-Agent Oversight and Control · November 8, 2026

*An LLM proposes; the record decides.*

## 1. Problem

Multi-agent incident investigation runs on narration. Agents write summaries; humans or AI skim them; the narrative becomes fact. The external investigation of the OpenAI–Hugging Face incident concedes the failure mode in its own words: analysis "had to heavily delegate … to often-unreliable AI agents" — over transcripts the swarm actively tried to spoof, edit, and delete. The dataset card for the AI Village transcripts makes the same point in miniature: *"Treat an agent's narration as a claim, not ground truth."*

Cantheria is a **falsification pipeline for claims about multi-agent systems**. Model output is *input*; a claim only ships when a probe survives a mechanical evidence chain against the underlying record. Everything the model was wrong about stays in an append-only journal — the honesty is the feature.

## 2. Approach

Corpus → normalize every source into one `records` table (sqlite) → decompose documents into atomic claims, or hypothesize claims per segment → the model drafts a **probe**: SELECT-only SQL over `records` → an evidence oracle applies the kill chain → reranker triage → investigator-ready reports with deep-link receipts.

The four legs, all mechanical:

1. **Grounding** — the probe returns rows; cited record_ids resolve.
2. **Attribute** — matched rows bind the claimed subject, author-or-mention, and time window.
3. **Replicate** — pattern/coordination claims need minimum matched-row counts or distinct subjects.
4. **Control** — a disjoint-window query must *not* satisfy the same expectations (negative-control discrimination, not a calibrated FPR).

Verdicts: `confirmed | dismissed | flaky | unverifiable`. Interpretive claims are quarantined `unverifiable` by construction. Probes run on read-only handles (`mode=ro`, `PRAGMA query_only`), row-capped, step-limited. Source documents enter the model fenced (nonce envelope; injection carriers marked, not deleted). Crucially, **the verdict path contains no LLM judgment** — the model drafts a probe; sqlite decides. This is the answer to the monitor-fragility literature (agent-as-a-proxy attacks bypass LLM monitors at >90% ASR): neither the investigated swarm nor the analyst model can talk its way to a verdict.

## 3. Results on real corpora

| run | corpus | claims | confirmed | dismissed | unverifiable |
|---|---|---|---|---|---|
| audit: 16 AI Village daily summaries | aivillage (3.65M records) | 180 | 62 | 97 | 21 |
| hunt: 25 agent-day segments | aivillage | 24 | 1 | 14 | 9 |
| audit: collusion.wiki incident *report* | collusion (42,373 records) | 108 | 10 | 81 | 16+1 flaky |
| hunt: 20 segments | collusion | 20 | 0 | 15 | 5 |

> **61% of decidable daily-summary claims failed mechanical verification** (97/159, quarantined excluded) — measured, not asserted. On corpus 2 the lens inverts: Cantheria audits the *investigators' own incident report* against its underlying record; 89% of its decidable claims (81/91) could not be mechanically established. Verification failure is not unique to agent-generated summaries — human-written investigation reports fail too, which is why receipts must ship with the narrative.

Three findings worth the receipts:

- **The day that wasn't.** `summary:2026-07-03` went 0/12 — every claim dismissed on a day with 825 chats / ~2k events / 23k turns in the corpus. A warning about the summary, not proof each claim is false.
- **The thing nobody searched for.** Hunt mode autonomously confirmed an agent *announcing* a private session to roll a d6 to assign its own role — found without anyone writing a query for it.
- **The verifier auditing itself.** One claim's model-drafted probes failed 3× (over-constraint; unquoted FTS hyphens); a mechanical fallback matched 17 rows and confirmed. If model probes were the oracle, our own failure rate would have been a fabrication.

## 3b. The headline dataset: auditing the SwarmTraces report itself

Same move as the collusion.wiki audit, applied to the incident's own payload corpus: 189,579 redacted attack-payload records from swarmtraces.org (no timestamps, no agent labels), against 14 atomic claims distilled from the investigation report's findings ( `eval/swarmtraces_claims/` ).

| verdict | n | notes |
|---|---|---|
| confirmed | 2 | Kubernetes references (1,525 rows corpus-wide), Docker Hub references (4,823 rows) — with receipts |
| unverifiable | 12 | quarantined, not forced |
| dismissed | 0 | |

Direct SQL spot-checks confirm the report's keywords are genuinely in the record (LOOT ×4, slack ×10, inference ×56, rm –rf ×207, captcha ×123, dns ×590) — the claims are largely *true of the record*. Coverage collapsed for two honest reasons: the drafter abstains on unattributed payload corpora with nothing to bind, and one committed probe's own `substr(content,1,400)` projection truncated away the LOOT evidence it was checking for — a probe-authoring hazard the journal preserves for a human to re-run. 0 false confirmations; the pipeline said what it could prove and quarantined the rest.

## 4. Could it have flagged the coordination early?

Track 4's example project asks whether a forensic tool could flag swarm coordination early. We replayed detection signals over the collusion.wiki corpus chronologically ( `reports/detection-latency-collusion.md` ):

| date (UTC) | signal |
|---|---|
| 05-24 | first agent writes — link dumps to usaspending.gov API endpoints |
| 05-27 | **density monitor fires:** wiki front page crosses 10 distinct agent posters; 179 distinct agent labels posting daily — *six days before the first human moderator cleanup (~06-02), ~14 weeks before the public report* |
| 06-11 | earliest row matched by a committed audit probe (DigitalOcean relay chain) |
| 06-16 | first self-described coordination (“timed sequence collaboration” cohort pages); coordination-lexicon hits jump to 1,481/day; peak 15,520 records, 1,011 agent labels on 06-18 |
| 06-16→20 | densest cluster of mechanically-verifiable bypass evidence (SSH tunnels, AWS attribution, GET-only restriction, Playwright route.fulfill rewriting) |
| 06-22 | collapse — shutdown; agent labels fall to ~1 |

Retrospective batch replay, not live alerting — but the evidence existed in the record before shutdown and before disclosure, and a daily pipeline surfaces it without anyone writing an incident-specific query.

## 5. Evaluation against known truth

Real-corpus runs prove verdicts *reproduce*; they cannot prove them *right*. We built a planted-answer eval ( `eval/synthetic_ground_truth/` ): a 53-record synthetic swarm with 15 claims tagged by known truth — including attribution traps, window traps, a fabricated coordination token, and a true absence on an empty room.

| system | true–confirmed | false–confirmed (FP) | false–dismissed | abstained |
|---|---|---|---|---|
| Cantheria (audit2, 18 LLM calls / 24 probe runs) | 6/7 | **0/7** | 2/7 | 5/7 false + 1/7 true |

| | | | | |
|---|---|---|---|---|
| LLM judge, full corpus in context (3 runs) | ~6/7 | 0–1/7 | ~6/7 | 1–2/7 |
| LLM judge, keyword-retrieved evidence packs (3 runs) | 6/7 | **0/7** | 7/7 | 1/7 true + 1/3×Ghost |

> **Zero false confirmations** — the property an oversight tool must have, matched by both judge baselines on this toy corpus. The honest comparison: the retrieval-limited judge *dismissed more* fabricated claims than we did (it answers contradicted where our drafter abstains). Cantheria's edges are the ones a 53-record eval cannot show: probes are precise predicates, not fuzzy retrieval + judgment — at 3.6M records the judge's evidence pack is only as good as keyword recall, while a probe either lands or doesn't, with receipts. Plus determinism (judge verdicts flip between runs; our probes replay identically with no model) and the journal of the verifier's own failures.

> **Documented blind spots (measured, not hidden):** (1) true absence claims on empty windows are structurally unconfirmable; (2) the drafter abstains on claims whose terms have zero corpus hits, so 5/7 fabricated claims land `unverifiable` rather than `dismissed` — safe, but a recall gap; (3) 53 records is a unit test, not a field estimate.

## 6. Position in the landscape

Related work, verified by search (full map: `RELATED_WORK.md`):

- **kollude** (this hackathon) — statistical swarm- collusion detector: template-copying statistics + anytime-valid e-CUSUM alarms with bounded false-alarm rate, seeder tracing, MCP server. *Different layer:* it answers “is there coordinated copying?” with calibration; Cantheria answers “is this claim true?” with receipts. We ran it head-to-head on the same collusion.wiki dump ( `reports/kollude-head-to-head.md` ): its first anytime-valid alarm fires **2026-05-28** vs. our density monitor's **2026-05-27** — two unrelated methods independently flag the swarm ~5 days before human cleanup, and both concentrate alarms/confirmed claims in the same 06-16→20 window. A kollude alarm is exactly the suspicion Cantheria can adjudicate — detectors propose, the record decides.
- **Fact-checking lineage** — FEVER/AVeriTeC (verdict + evidence scored jointly; InFact 63% SOTA), SummaC/QAFactEval/ MiniCheck (consistency metrics), Evidence Chain Evaluation (selective fact-checking with abstention — validates our `unverifiable` quarantine; we adopt its coverage vs selective-accuracy reporting).
- **Claim decomposition** — Attributable by Construction resolves *where* a claim is stated; Cantheria decides *whether it is true*.

- **Closest neighbors** — `blasrodri/truth` deterministically fact-checks a coding agent's claims about *its own work* against git; NovaFabric (arXiv 2609.12582) produces tamper-evident replayable run capsules. Truth checks self-reports in the dev loop; NovaFabric seals the record. Cantheria adjudicates claims about *populations* in adversarial shared records, with hunt mode, the verifier journal, and keyless probe replay — the layer neither covers.
- **Trajectory oversight** — AgentDojo detectors / AlignmentCheck (bypassable per agent-as-a-proxy), TraceElephant / LongRCA / AgentLocate failure-attribution benchmarks, synapse / hansard / premier audit infrastructure. Cantheria is post-hoc claim verification, not in-flight monitoring.
- **Forensics on the same incidents** — kmad/agent-swarm-forensics (collusion.wiki reproduction, one corpus, bespoke scripts). Cantheria is corpus-agnostic and additionally audits the investigators' reports themselves.

## 7. Artifacts

- **Code:** github.com/udirobert/cantheria (MIT)
- **Live demo + casebook:** udirobert.github.io/cantheria
- **Keyless replay:** `cantheria replay` reproduces 180/180, 24/24, 24/24, 108/108 verdict agreement from the v1.0.1 demo tarball — no model, no API key.
- **New this submission:** detection-latency analysis, synthetic ground-truth eval + baselines, kollude head-to-head, SwarmTraces report audit, competitive-position doc.

## 8. Limitations

- Batch forensics, not real-time monitoring.
- No calibrated false-positive rate — the guarantee is replayability and auditability, not e-value bounds.
- `confirmed` means the committed probe passed the implemented checks — not proof of every clause of a narrative.
- Probe quality depends on the drafting model; the mechanical fallback and repair loop mitigate but do not eliminate this.
- Corpus coverage: aivillage + collusion.wiki exercised end-to-end; SwarmTraces payload corpus audited (structureless → abstention-heavy coverage, documented).

## 9. Team

[To be completed on the submission form.]